



Contrastive Language Image Pretraining (CLIP)

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OUTLINE

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Introduction to CLIP – What it is and why it is important?

Contrastive Learning in CLIP – How CLIP learns from paired image-text data?

CLIP's Architecture and Training – VLM components and training

Limitations and Challenges – Key challenges and limitations in CLIP's approach

CONTRASTIVE LANGUAGE-IMAGE PRETRAINING



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vision, speech, text, etc

1. **Multimodal:** vision + language (text)

2. **Internet Scale**

- Trained on image-caption pairs
- Broad visual understanding

3. **Zero-Shot Capabilities**

- No training data needed for many classification tasks

* Earlier search engines use alt text to identify images as it couldn't interpret earlier.

training data

source of training data



Man wearing blue mask



Colorful buttons



Hot air balloon in foggy mountains



Peach fruit on a tree

CLIP DATASET & SCALE



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Web ImageText (WIT) Dataset

- 400M image-text pairs gathered from the internet
- Derived from alt text or captions
- Similar word-count to used in GPT-2
- Not publicly available

Training Time

- Largest ResNet-based (RN50x64) CLIP trained in ~18 days on 592 V100 GPUs
- Largest ViT-based CLIP in ~12 days on 256 V100 GPUs.

Two different
architectures

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CONTRASTIVE LEARNING



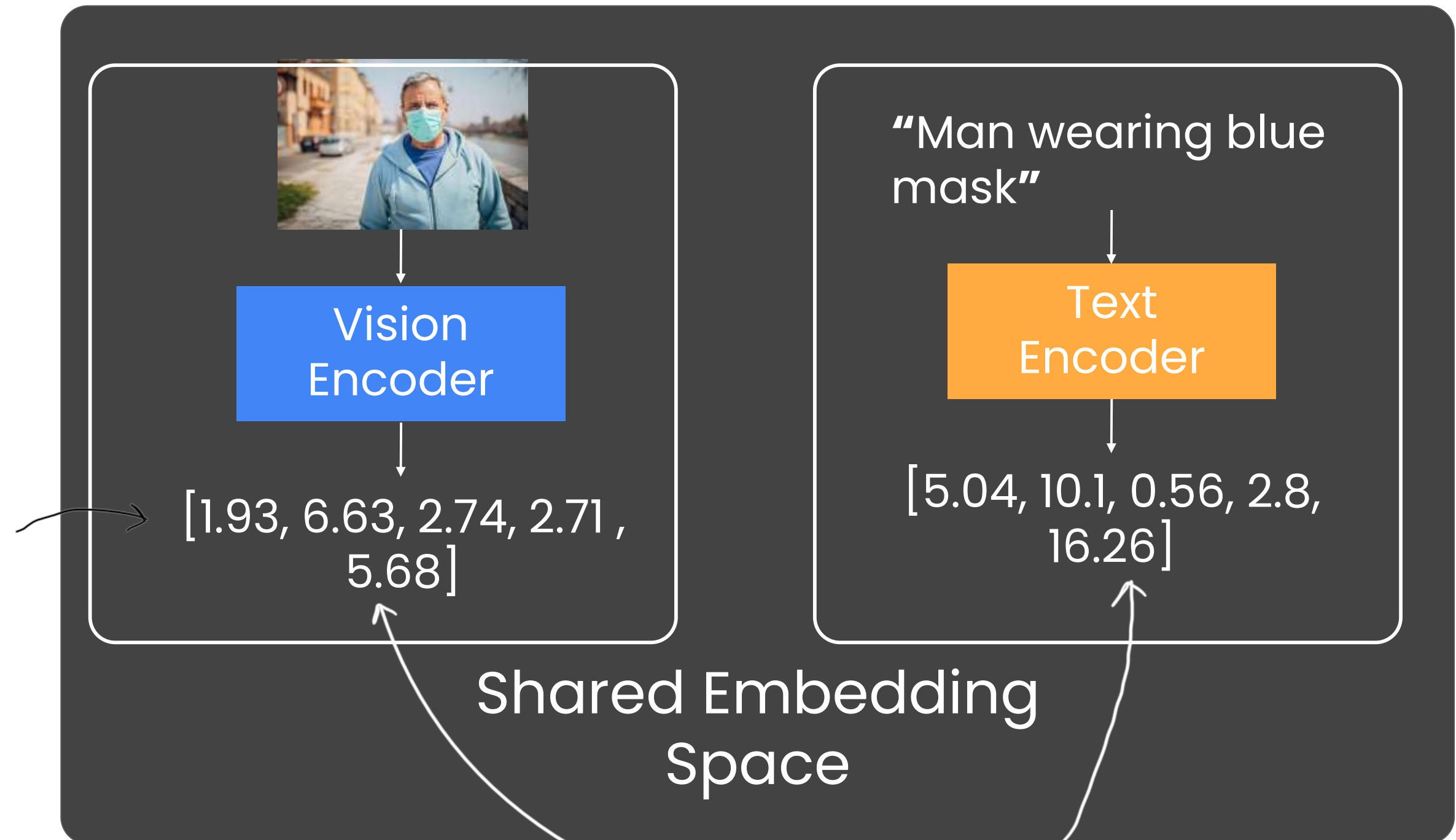
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1. Encode image as vector
2. Encode text as same sized vector

What determines
this → 512 numbers in a
vector usually around

Embedding:

A vector representation of some data (text, image, audio) usually reflecting semantic meaning



* In a shared embedding space as both are same size

CONTRASTIVE LEARNING

HIGH SIMILARITY FOR CORRECT PAIR



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Vision Encoder

[1.93, 6.63, 2.74, 2.71, 5.68]

“Man wearing blue
mask”

Text Encoder

[5.04, 10.1, 0.56, 2.8, 16.26]

Cosine Similarity

0.91

cosine
angle

$$\text{cosine similarity}(a, b) = \frac{a \cdot b}{\|a\| \|b\|}$$

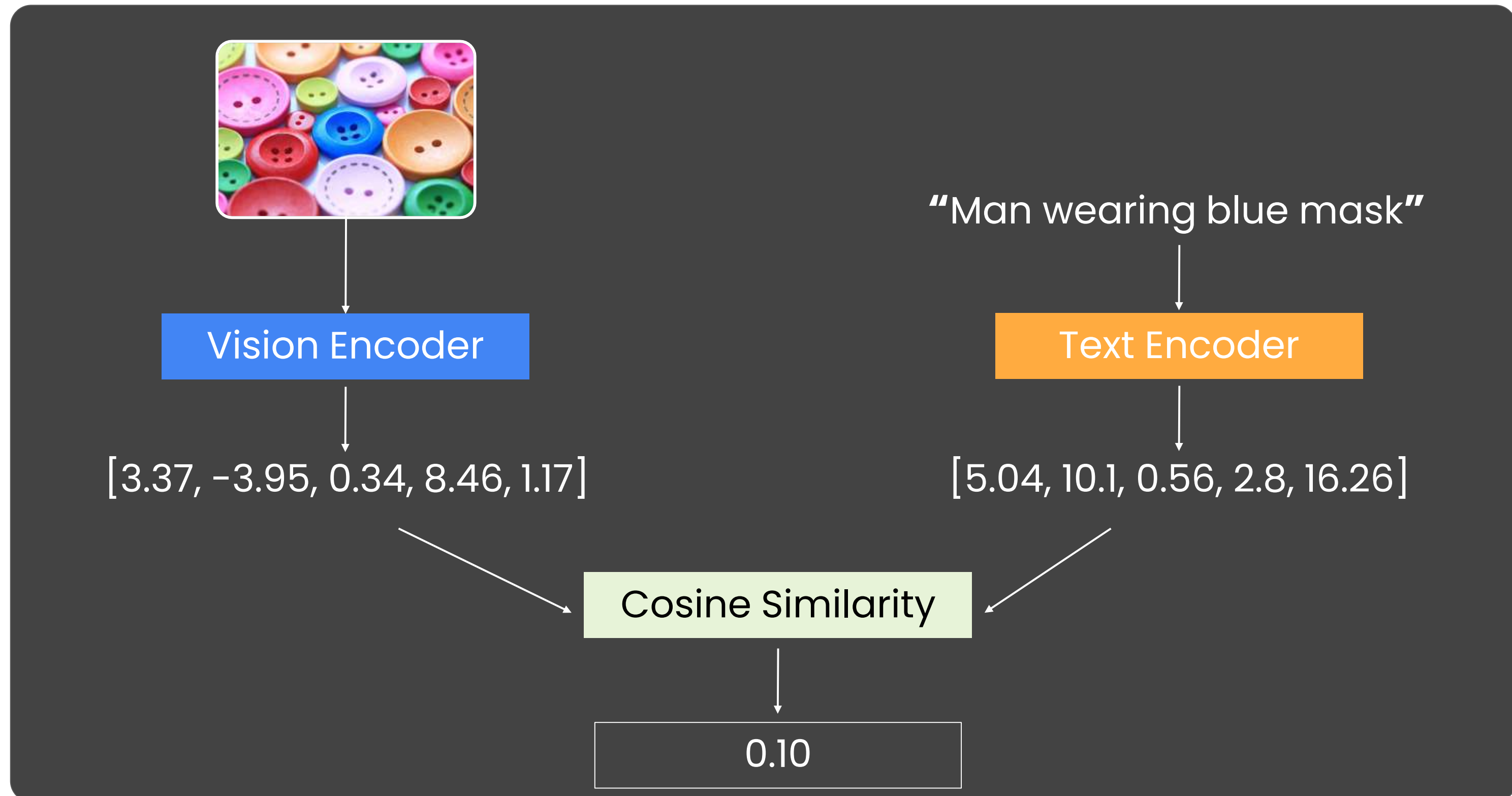
To understand two things are similar or not,

* when two vectors aligned, angle = 0°
cosine = 1

CONTRASTIVE LEARNING

LOW SIMILARITY FOR PAIR

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CONTRASTIVE LEARNING: EMBEDDING

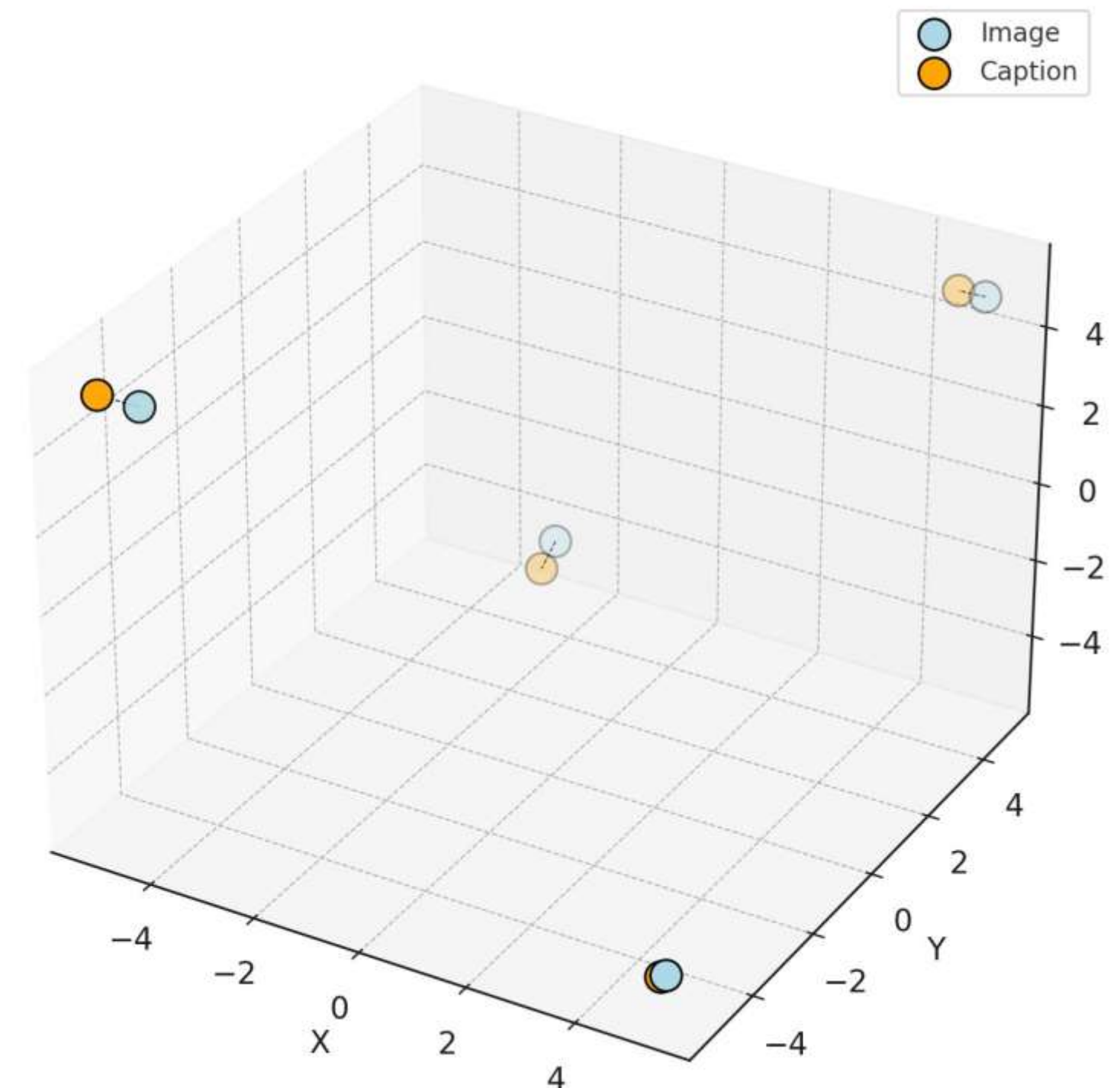


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Batch size of N image-text pairs

- **Image-text similarity:** Maximize similarity between an image and its caption while minimize its similarity between the image and other captions.
- **Text-image similarity:** Maximize similarity between a caption and its image while minimize the similarity between the caption and other images.

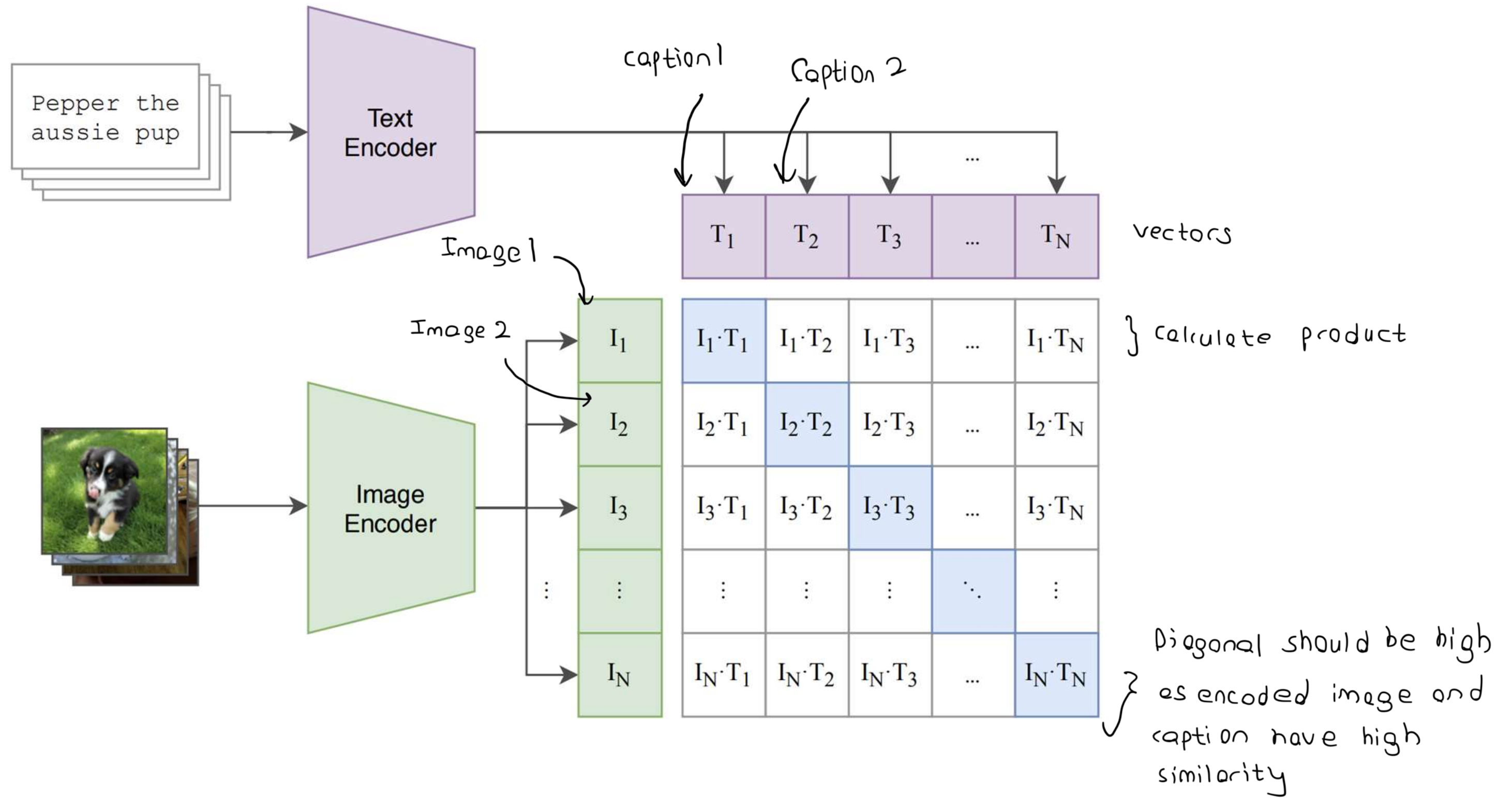
Four pairs of 3D points (Image and Caption)



CONTRASTIVE LEARNING: TRAINING LOSS



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CONTRASTIVE LEARNING: TRAINING LOSS

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* Symmetric contrastive Loss

guide to train the NN

$$\mathcal{L}_{\text{CLIP}} = \frac{1}{2} \left[\underbrace{-\frac{1}{N} \sum_{i=1}^N \log \frac{\exp(\text{sim}(\overbrace{I_i, T_i}^{\text{same image-caption pairs}}) / \tau)}{\sum_{j=1}^N \exp(\text{sim}(I_i, T_j) / \tau)}}_{\text{image-to-text loss}} + \underbrace{-\frac{1}{N} \sum_{i=1}^N \log \frac{\exp(\text{sim}(T_i, I_i) / \tau)}{\sum_{j=1}^N \exp(\text{sim}(T_i, I_j) / \tau)}}_{\text{text-to-image loss}} \right]$$

feedback setting to change weights and biases

* loss function is important in training the any NN/model

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CLIP ARCHITECTURE



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Vision Encoder

Two options

- ResNet-50 convolutional network or a vision transformer
- Output vector of size "512"

Text Encoder

- 12-layer transformer (with ~63 million parameters, 8 attention heads) similar in architecture to GPT-2
- Output vector size "512"

How adjusted? *Using loss function*

Parameters of both encoders are adjusted so that matching image/text pairs have higher similarity scores than mismatches

CLIP Architecture and Training



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Training

- No pre-trained weights from ImageNet or NLP models.
- Large mini batch sizes were used – on the order of 32,768 examples per batch – enabled by distributed training across many GPUs.

Contrastive Loss Choice

- Predicting captions word-for-word (a generative approach) was much slower to learn visual features.
- A contrastive loss was much faster.

* Create caption from image, then compare captions vs ground truth } Time consuming ∴ Contrastive learning taken using embedding alignment.

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CLIP LIMITATIONS



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not good

Abstract and Fine-Grained Tasks

- Struggles with counting objects or determining distances, which require more reasoning.
- Has difficulty distinguishing very similar subcategories (e.g., bird species or car models) in zero-shot settings.

Out-of-Distribution Generalization Gaps

- Performs poorly on specialized data like handwritten digits (e.g., MNIST), indicating limited coverage.
- Sensitive to prompt phrasing. Exact wording of input prompt can affect model predictions

CLIP LIMITATIONS



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Biases from Uncurated Web Data

- Absorbs societal biases present in image–caption pairs, leading to potentially harmful outcomes.
- Misclassification can be aggravated by incomplete label sets.

Ethical and Practical Concerns

- Could be misused for surveillance or facial recognition, raising privacy and ethical issues.
- Its open-ended design makes failures less predictable, highlighting the need for cautious deployment.



MODEL CARD

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instruct models used for chatting

Name	“openai/clip-vit-base-patch32”
Summary	Dual-encoder vision-language model (ViT-B/32 image encoder + Transformer text encoder) trained contrastively on ~400 M image-text pairs to enable zero-shot recognition, retrieval and embedding tasks.
Parameters	151 M
Release Date	January 2021
Developer	OpenAI
License	Code — MIT License; Weights released for research only under OpenAI terms (“no un-evaluated commercial deployment”)



THANK YOU

